

MANDAR DUMBRE

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EDUCATION

Bachelor of Technology, Electronics and Telecommunication Engineering

Nov 2022 - Present

Sardar Patel Institute of Technology

Relevant coursework: Computer Networks, Data Structures & Algorithms, DBMS, OOP, Operating Systems, SDLC

PROJECTS

GitChat AI – Repository Analyzer | Python, FastAPI, JavaScript, Tailwind CSS, OAuth2, JWT, Redis, Llama3:70B

- Engineered a FastAPI-based microservice providing AI-driven insights into public GitHub repositories
- Leveraged LangChain and HuggingFace Embeddings with FAISS to enable local, semantic code search
- Implemented a smart document ranking pipeline with multi-stage retrieval, chunking, and re-ranking via Groq API
- Engineered a JWT-secured, rate-limited backend enabling 1000+ indexed scans with 90%+ response relevance
- Achieved 30% faster analysis with improved maintainability and token-aware chunking for context preservation

Capital Compass – Portfolio Optimization Platform | Python, SQLite, PyPortfolioOpt, yfinance, Llama3:8B

- Built a real-time portfolio optimizer using Modern Portfolio Theory, custom strategies on USA & Indian equities
- Simulated future outcomes using Monte Carlo analysis & key metrics like VaR, Sharpe, Sortino, and Calmar
- Used Llama3:8B to generate natural language insights from historical performance for strategy explainability

FinWise – AI-Powered Personal Finance Dashboard | Python, Streamlit, Pandas, NumPy, Plotly, Groq API

- Developed a smart finance dashboard with categorization, budgeting, and anomaly detection using Z-score
- Delivered dynamic analytics, and AI-driven insights with an interactive UI using Streamlit and Plotly

Student Council Website | HTML, CSS, JavaScript, Bootstrap

- Built and deployed a mobile-responsive website to streamline event visibility, team structure, and outreach
- Enhanced usability and information flow for 1200+ students through intuitive navigation and modular design

TECHNICAL SKILLS

Languages & Frameworks: Java, Python, JavaScript, TypeScript, React.js, Node.js, Spring Boot

Web and App Development: HTML5, CSS3, Bootstrap, jQuery, REST APIs, MVC Architecture, JWT, OAuth

Databases and Tools: MySQL, PostgreSQL, MongoDB, Git, GitHub, VS Code, IntelliJ, Postman, Maven, Ant

EXPERIENCE

SP Jain Institute of Management & Research: Research & Data Mining Intern

Jan 2025 – Jun 2025

- Built a scalable scraper to extract 30K+ TripAdvisor reviews using Python, BeautifulSoup, and anti-bot logic
- Processed and analyzed user data with NumPy and Pandas to uncover patterns in digital behavior

DeepCytes Cyber Labs (UK): Cyber Intelligence Intern

Jun 2024 – Dec 2024

- Managed pipeline for 40+ cybersecurity posts, showcasing analytical and communication skills
- Curated 3+ case study decks to simplify complex cyber risks for stakeholders, enhancing data-backed storytelling

POSITIONS OF RESPONSIBILITY

Students' Council: Technical Secretary

Aug 2024 – July 2025

- Led coordination among 10+ teams for 15+ successful events, benefiting 1,200+ students
- Streamlined communication and logistics, improving event planning efficiency by 30%

Aelaan-E-Jung, Oculus: Head of Finance

Aug 2023 – July 2024

- Managed Rs 1L+ budget and secured Rs 30K in sponsorships across 12+ college-wide events
- Built reporting systems ensuring 100% transparency and real-time tracking of funds.

ACHIEVEMENTS

Published IEEE Author: IoT-based Prosthetic Arm Controlled by Human Gestures, ESCI 2025, Pune, India

Winner, Runners Up, 2nd Runners Up: At 6+ National and Institute level Hackathons and Case Study Competitions